Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/08/2004
PI Name:	Walker, Thad G Ph.D.		
Project Title:	All-optical high density atom sources		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	tgwalker@wisc.edu	**Fax:**	FY 608-265-2334
PI Organization Type:	UNIVERSITY	**Phone:**	608-262-4093
Organization Name:	University of Wisconsin-Madison		
PI Address 1:	Department of Physics		
PI Address 2:	1150 University Ave.		
PI Web Page:	http://www-atoms.physics.wisc.edu/		
City:	Madison	**State:**	WI
Zip Code:	53706	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	04/01/2001	**End Date:**	03/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
Contact Email:	lute.maleki@jpl.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and uncover mysteries of the universe obscured by Earth"s gravity		

Our approach to producing high density, high phase space density atoms is to laser cool atoms confined in a holographically produced far-off-resonance lattice. This Holographic Atom Trap (HAT) is produced by imaging a hologram onto a cloud of laser-cooled atoms. The HAT produces strongly confining forces on atoms in a 10-100 micron volume. Once in the HAT, the atoms will be further cooled using lasers to temperatures of a few micro-Kelvin.
The ultimate densities and phase space densities that can be reached in the HAT depend on the collisional properties of the trap and heating due to photon scattering. Both effects are minimized in the HAT by optical pumping into dark states. We will investigate the possibility of direct laser cooling to quantum degeneracy in the HAT.

We seek to use the HAT as a source for studying quantum transport through atom waveguides. Our waveguide will be formed by a laser beam with a hole down the middle. Characteristic steps in the transmission of the guide as a function

Task Description:	of energy are quantized in units of the DeBroglie wavelength. The small energy scales involved mean that gravity is a substantial complication, and later manifestations of the experiment will be substantially improved in a microgravity environment. We seek to develop new high density, high phase space density sources of cold atoms for scientific exploitation of microgravity environments. Such sources potentially have inherent advantages as compared to currently used technology, namely speed, atom efficiency, and power consumption. We propose to develop and study the properties of these sources, and to use them to address a fundamental physics problem ideally suited to microgravity: quantization of atom conductance through an atom waveguide. Development of high-density atom sources by laser cooling will have advantages over current technology in terms of speed, efficiency, and power consumption. These new sources may be of considerable interest to a wide variety of experiments using ultracold atoms on the ground and in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This year we have succeeded in producing the highest density trapped atoms that exist outside of Bose condensates. The method is similar to that normally used for condensation, namely using evaporative cooling to increase phase space density. In an optical trap that normally comes at the cost of decreasing the atom density. However, after evaporation we recompress the optical trap adiabatically, preserving the phase space density and giving an order-of-magnitude increase in the atom density. In this manner we have demonstrated densities in excess of 10^15 atoms/cc. Furthermore, the cold clouds so obtained are extremely small, 8 microns or less, and therefore can be used for investigating the very interesting dipole blockade phenomena that is predicted to be a robust new paradigm for quantum entanglement.
Bibliography Type:	Description: (Last Updated: 11/10/2009)
Articles in Peer-reviewed Journals	R. Newell, J. Sebby, and T. G. Walker "Dense Atom Clouds in a Holographic Atom Trap" ", Opt. Lett. 28, 1266 (2003) , Sep-2003
Articles in Peer-reviewed Journals	M. Saffman and T. G. Walker "Entangling single and N atom qubits for fast quantum state detection and transmission" submitted for publication Feb 2004. , Feb-2004